Introduction:

Online shopping is now a big part of our daily lives. It allows people to buy products from home without going to physical stores. Since many people use online stores, it is important to have a well-designed database to manage products details, customers information, orders, and payments. The system should ensure that all operations, including adding products, placing orders, and processing payments, are handled efficiently and accurately. It aims to simplify management for administrators and enhance the shopping experience for customers.

Project Goals and Objectives:

Goals:

1. Efficient management of customer information and accounts.
2. Accurate and reliable order processing.
3. Secure and smooth payment processing.
4. Proper inventory and product data management.
5. A user-friendly interface for both administrators and customers.
6. Well-organized and easily retrievable data.
7. Consistent and high-performing system operation.

Objectives:

- Maintain accuracy and reliability of stored customer and order data.
- Reduce manual errors in order and payment processes.
- Ensure secure online payment transactions.
- Improve data retrieval speed for searches and reports.
- Keep data relationships consistent through key constraints.

Entities, Attributes, and Constraints:

1. Website

Attributes:

- Domain_Name (Primary Key)
- Customer_ID (Foreign Key)

Constraints:

- Domain_Name must be unique.
- Customer_ID must exist in the Customer table.

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2. Customer

Attributes:

- Customer_ID (Primary Key)
- Name (First_Name, Last_Name)
- Address (City, State, ZIP)
- Username (Unique)
- User_Password

Constraints:

- Customer_ID must be unique.
- Username cannot be duplicated.
- All attributes cannot be null.
- Name should be a composite attribute
- The **Username** must be unique.

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3. Product

Attributes:

- Product_ID (Primary Key)
- Product_Name
- Product_Category
- Product_Price
- Product_Availability
- Order_ID (Foreign Key)

Constraints:

- Product_ID must be unique.
- Product_Price must be greater than 0.
- Order_ID must exist in the Order table.

- All attributes cannot be null.

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4. Order

Attributes:

- Order_ID (Primary Key)
- Order_Date
- Total_Price
- Customer_ID (Foreign Key)

Constraints:

- Order_ID must be unique.
- Customer_ID must exist in the Customer table.
- All attributes cannot be null .

- Order_Date inserted by default

- The **Order_Date** defaults to the current date when a new record is created.

5. Tracking Information

Attributes:

- Tracking_ID (Primary Key)
- Courier
- Order_ID (Foreign Key)

Constraints:

- Tracking_ID must be unique.
- Each order can have only one tracking record.
- All attributes cannot be null

6. Payment Details

Attributes:

- Payment_ID (Primary Key)
- Payment_Date
- Payment_Type (Card, Cash on Delivery)
- Customer_ID (Foreign Key)

Constraints:

- Payment_ID must be unique.
- All attributes cannot be null.
- Customer_ID must exist in the Customer table.
- "CASH" sub-attribute is chosen by default in case nothing been chosen in " Payment_Type " composite attribute

7. Cart

Attributes:

- Cart_ID (Primary Key)
- Customer_ID (Foreign Key)
- Product_ID (Foreign Key)
- Quantity

Constraints:

- Cart_ID must be unique.
- Quantity must be greater than 0.
- Customer_ID and Product_ID must exist in their respective tables.
- The **Product_ID** must exist in the Product table as a foreign key.
- The **Quantity** must be greater than 0.

Scenario for the Online Shopping Database Schema:

An e-commerce company operates an online shopping platform where customers can browse products, add items to their shopping cart, place orders, and track their shipments. The system stores and manages all required data using the tables shown in the schema.

1. Customer Uses the Website

Each Customer registers on the Website to start shopping.

The Website keeps a record of the domain name and links each account to a Customer_ID to indicate which customers use the website.

A customer provides personal information such as:

- First and last name
- Address (city, state, ZIP)
- Username and password

This information is stored in the CUSTOMER table.

2. Browsing and Adding Products to the Cart

The PRODUCT table stores all available items, including:

- Product name
- Category
- Price
- Availability

As customers browse the website, they can choose products to buy.

Each time a customer adds an item to the shopping cart, a record is inserted into the CART table, capturing:

- The selected Product_ID
- The Customer_ID who added it
- The quantity chosen

This means the CART table represents each product–customer pairing and the number of units the customer intends to purchase.

3. Order Placement

When the customer proceeds to checkout, the system creates an entry in the

ORDER table.

Each order includes:

- A unique Order_ID
- The order date
- Total price
- The Customer_ID who placed the order

The order may contain multiple products.

The relationship between ORDER and PRODUCT (through ORDER_ID in PRODUCT) allows the system to list which products were included in each order.

4. Payment Processing

After placing an order, the customer must complete payment.

The PAYMENT_DETAILS table stores:

- Payment ID
- Payment date
- Payment type (cash, card)
- The associated Customer_ID

This ensures that each customer can have multiple payments recorded in the system, depending on their orders.

5. Shipment and Tracking

Once payment is confirmed, the order is processed for shipping.

The TRACKING_INFO table stores delivery and shipment details:

- A unique tracking number
- Courier name
- The related Order_ID

Customers can use this tracking number to monitor the shipping progress of their order.

Summary:

1. A customer registers and logs into the website.

2. The customer browses available products.

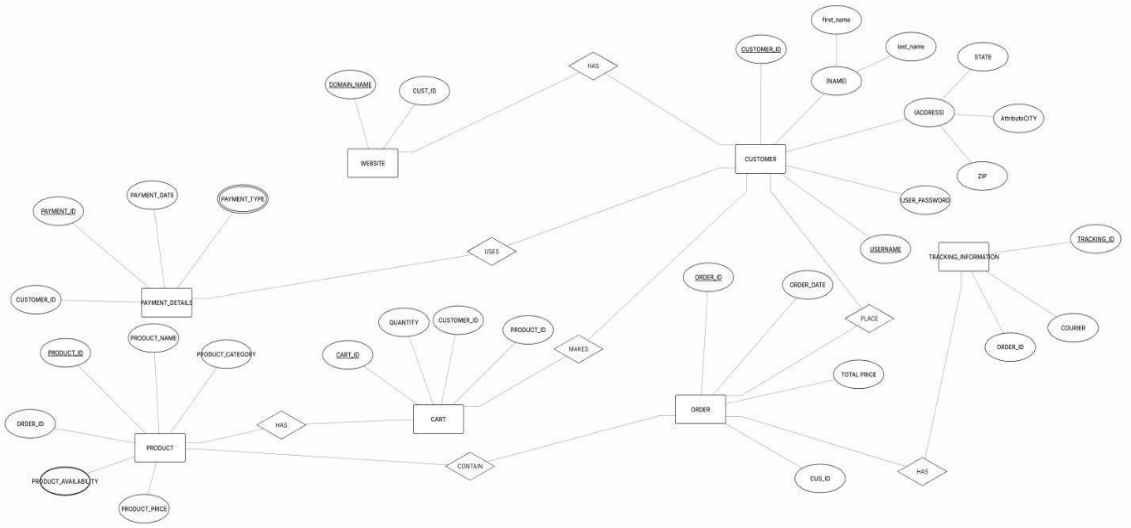
3. They add selected items to their cart.

4. The customer places an order, which stores product details and pricing.

5. Payment information is recorded in the Payment Details table.

6. The system assigns tracking information so the customer can monitor the delivery.

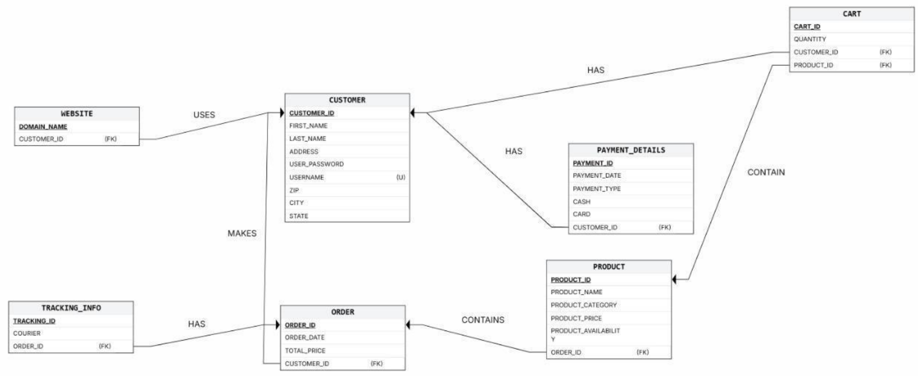
I. ER digram:



I. Relational schema:

The schema is composed of 7 tables: Website, Customers, Payment details, Products, Order, and Tracking info, cart.

Using key constraints, the schema connects all of the tables and organizes and structures the data from the ER diagram. It helps user interaction with the database.



Relationships

1. Website > Customer (1:N)
2. Customer > Order (1:N)
3. Customer > Payment Details (1:1)
4. Order > Tracking Information (1:1)
5. Product > Order (1:N)
6. Customer > Cart (1:1)
7. Cart > Product (N:1)

II. Attributes table and their types of constraints:

1.Website

Attribute	DataType/Size	Constraint
Domain_Name	VARCHAR (255)	(PK)
Customer_ID	INT	(FK)

2.Customer

Attribute	Sub-Attribute	DataType/Size	Constraint
Customer_ID (PK)		INT	(PK)
Name	First_Name	VARCHAR (255)	cannot be null
	Last_Name	VARCHAR (255)	cannot be null
Address	City	VARCHAR (50)	cannot be null
	State	VARCHAR (50)	cannot be null
	ZIP	CHAR (6)	cannot be null
Username		VARCHAR (30)	must be unique
User_Password		VARCHAR (255)	cannot be null

3.Product

Attribute	DataType/Size	Constraint
Product_ID	INT	PK
Product_Name	VARCHAR (255)	cannot be null
Product_Category	VARCHAR (255)	cannot be null
Product_Price	FLOAT	Check Cons. Must be greater than 0
Product_Availability	INT	cannot be null
Order_ID	INT	FK: must exist in the ORDER table

4.Order

Attribute	DataType/Size	Constraint
Order_ID (PK)	INT	PK: must be unique
Order_Date	DATE	Default Constraint
Total_Price	FLOAT	cannot be null
Customer_ID (FK)	INT	FK: must exist in the Customer table

5.Tracking_Info

Attribute	DataType/Size	Constraint
Tracking_ID (PK)	INT	PK: must be unique
Courier	VARCHAR (255)	cannot be null
Order_ID (FK)	INT	FK: must exist in the ORDER table, Trigger: there is one tracking info per order

6.Payment_Details

Attribute	Sub-Attribute	DataType/Size	Constraint
Payment_ID (PK)		INT	(PK)
Payment_Date		DATE	cannot be null
Payment_Type	CASH	VARCHAR (10)	cannot be null, Default Cons.
	CARD	INT	cannot be null
Customer_ID (FK)			FK: must exist in the Customer table

7.Cart

Attribute	DataType/Size	Constraint
Cart_ID (PK)	INT	PK: must be unique
Customer_ID (FK)	INT	FK: must exist in the Customer table
Product_ID (FK)	INT	FK:must exist in the Product table
Quantity	INT	Check Con.: must be greater than 0

III. SQL:

DDL:

```
CREATE TABLE CUSTOMER(
  CUSTOMER_ID INT NOT NULL,
  FIRST_NAME VARCHAR(255) NOT NULL,
  LAST_NAME VARCHAR(255) NOT NULL,
  ADDRESS VARCHAR(255) NOT NULL,
  USER_PASSWORD VARCHAR(255) NOT NULL,
  USERNAME VARCHAR(30) NOT NULL,
  ZIP CHAR(6) NOT NULL,
  CITY VARCHAR(50) NOT NULL,
  STATE VARCHAR(50) NOT NULL,
```

```
  PRIMARY KEY (CUSTOMER_ID),
  UNIQUE (CUSTOMER_ID),
  UNIQUE (USERNAME));

CREATE TABLE WEBSITE(
  DOMAIN_NAME VARCHAR NOT NULL,
  CUSTOMER_ID INT NOT NULL,
  PRIMARY KEY (DOMAIN_NAME),
  FOREIGN KEY (CUSTOMER_ID) REFERENCES CUSTOMER(CUSTOMER_ID),
  UNIQUE (DOMAIN_NAME));

CREATE TABLE ORDER(
  ORDER_ID INT NOT NULL,
  ORDER_DATE DATE NOT NULL,
  TOTAL_PRICE FLOAT(4,6) NOT NULL,
  CUSTOMER_ID INT NOT NULL,
  PRIMARY KEY (ORDER_ID),
  FOREIGN KEY (CUSTOMER_ID) REFERENCES CUSTOMER(CUSTOMER_ID));

CREATE TABLE PRODUCT(
  PRODUCT_ID INT NOT NULL,
  PRODUCT_NAME VARCHAR(255) NOT NULL,
  PRODUCT_CATEGORY VARCHAR(255) NOT NULL,
```

```
  PRODUCT_PRICE FLOAT(4,6) NOT NULL,
  PRODUCT_AVAILABILITY INT NOT NULL,
  ORDER_ID INT NOT NULL,
  PRIMARY KEY (PRODUCT_ID),
  FOREIGN KEY (ORDER_ID) REFERENCES ORDER(ORDER_ID));

CREATE TABLE TRACKING_INFO(
  TRACKING_ID INT NOT NULL,
  COURIER VARCHAR(255) NOT NULL,
  ORDER_ID INT NOT NULL,
  PRIMARY KEY (TRACKING_ID),
  FOREIGN KEY (ORDER_ID) REFERENCES ORDER(ORDER_ID));
CREATE OR REPLACE TRIGGER prevent_duplicate_tracking
BEFORE INSERT ON TRACKING_INFO
FOR EACH ROW
DECLARE
    existing_count INT;
BEGIN
    SELECT COUNT(*) INTO existing_count
    FROM TRACKING_INFO
    WHERE ORDER_ID = :NEW.ORDER_ID;

    IF existing_count > 0 THEN
```

```
        RAISE_APPLICATION_ERROR(-20001, 'Each order can only have one tracking
information');
    END IF;
END;
CREATE TABLE PAYMENT_DETAILS(
  PAYMENT_ID INT NOT NULL,
  PAYMENT_DATE DATE NOT NULL,
  PAYMENT_TYPE VARCHAR(255) NOT NULL,
  CASH VARCHAR(10) NOT NULL,
  CARD INT NOT NULL,
  CUSTOMER_ID INT NOT NULL,
  PRIMARY KEY (PAYMENT_ID),
  FOREIGN KEY (CUSTOMER_ID) REFERENCES CUSTOMER(CUSTOMER_ID));

CREATE TABLE CART(
  CART_ID INT NOT NULL,
  QUANTITY INT NOT NULL,
  CUSTOMER_ID INT NOT NULL,
  PRODUCT_ID INT NOT NULL,
  PRIMARY KEY (CART_ID),
  FOREIGN KEY (CUSTOMER_ID) REFERENCES CUSTOMER(CUSTOMER_ID),
  FOREIGN KEY (PRODUCT_ID) REFERENCES PRODUCT(PRODUCT_ID));
```

Creating tables on Oracle:

```sql
CREATE TABLE CUSTOMER (
  CUSTOMER_ID INT NOT NULL PRIMARY KEY,
  FIRST_NAME VARCHAR(255) NOT NULL,
  LAST_NAME VARCHAR(255) NOT NULL,
  ADDRESS VARCHAR(255) NOT NULL,
  USER_PASSWORD VARCHAR(255) NOT NULL,
  USERNAME VARCHAR(30) NOT NULL UNIQUE,
  ZIP CHAR(6) NOT NULL,
  CITY VARCHAR(50) NOT NULL,
  STATE VARCHAR(50) NOT NULL
);
```

```
Table created.
```

```sql
CREATE TABLE WEBSITE (
  DOMAIN_NAME VARCHAR(255) NOT NULL PRIMARY KEY,
  CUSTOMER_ID INT NOT NULL,
  FOREIGN KEY (CUSTOMER_ID) REFERENCES CUSTOMER(CUSTOMER_ID)
);
```

```
Table created.
```

```sql
CREATE TABLE ORDERS (
  ORDER_ID INT NOT NULL PRIMARY KEY,
  ORDER_DATE DATE DEFAULT CURRENT_TIMESTAMP,
  TOTAL_PRICE float (10) NOT NULL,
  CUSTOMER_ID INT NOT NULL,
  FOREIGN KEY (CUSTOMER_ID) REFERENCES CUSTOMER(CUSTOMER_ID)
);
```

```
Table created.
```

```sql
CREATE TABLE PRODUCT (
  PRODUCT_ID INT NOT NULL PRIMARY KEY,
  PRODUCT_NAME VARCHAR(255) NOT NULL,
  PRODUCT_CATEGORY VARCHAR(255) NOT NULL,
  PRODUCT_PRICE DECIMAL(10, 2) NOT NULL CHECK (PRODUCT_PRICE > 0),
  PRODUCT_AVAILABILITY INT NOT NULL,
  ORDER_ID INT NOT NULL,
  FOREIGN KEY (ORDER_ID) REFERENCES ORDERS(ORDER_ID)
);
```

```
Table created.
```

```sql

CREATE TABLE TRACKING_INFO (
  TRACKING_ID INT NOT NULL PRIMARY KEY,
  COURIER VARCHAR(255) NOT NULL,
  ORDER_ID INT NOT NULL,
  FOREIGN KEY (ORDER_ID) REFERENCES ORDERS(ORDER_ID),
  UNIQUE (ORDER_ID)
);
```

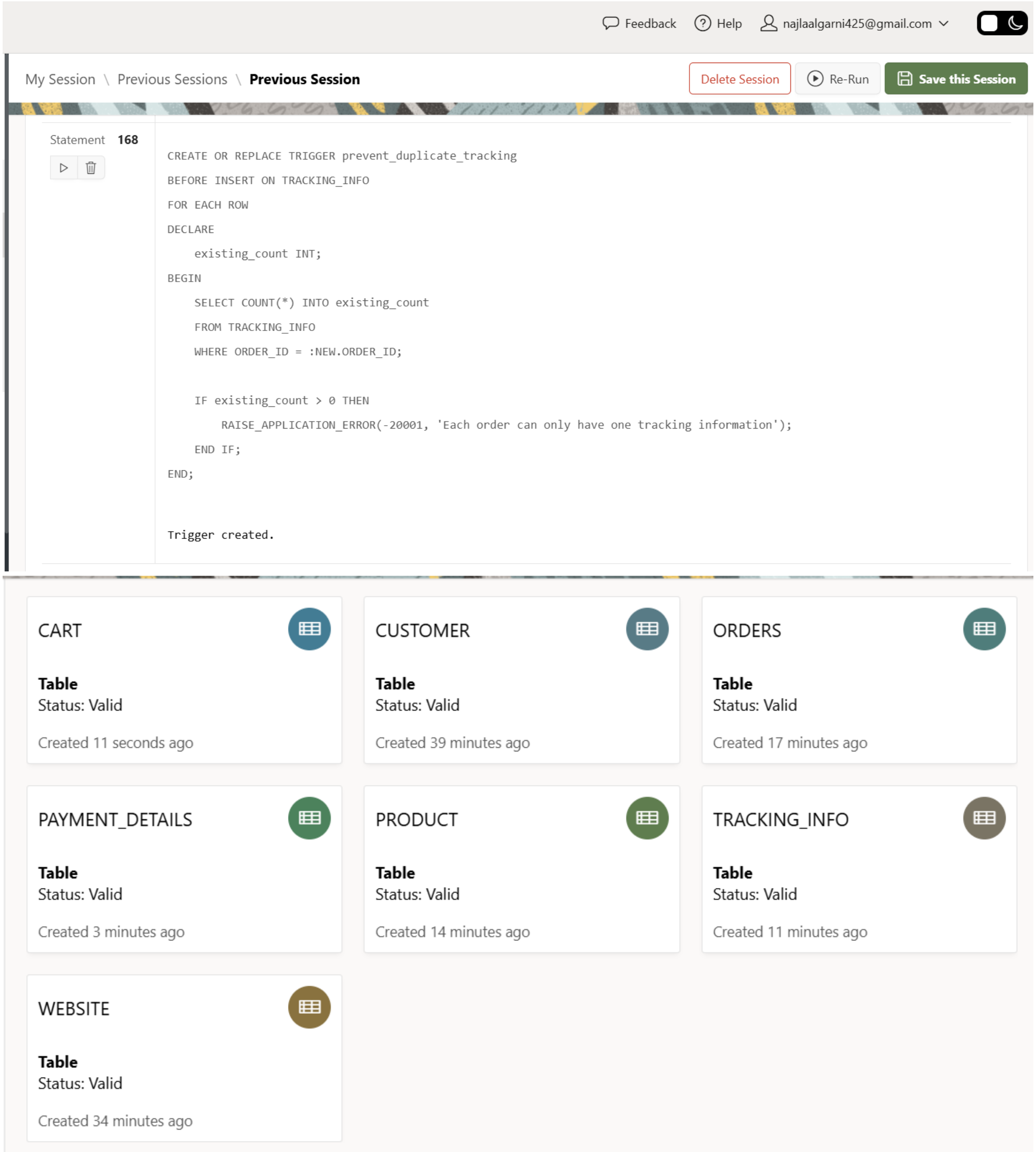
```
Table created.
```

```
CREATE TABLE PAYMENT_DETAILS(
  PAYMENT_ID INT NOT NULL,
  PAYMENT_DATE DATE NOT NULL,
  PAYMENT_TYPE VARCHAR(255) NOT NULL,
  CASH VARCHAR(10) NOT NULL,
  CARD INT NOT NULL,
  CUSTOMER_ID INT NOT NULL,
  PRIMARY KEY (PAYMENT_ID),
  FOREIGN KEY (CUSTOMER_ID) REFERENCES CUSTOMER(CUSTOMER_ID));
```

```
able created.
```

```
CREATE TABLE CART (
  CART_ID INT NOT NULL PRIMARY KEY,
  QUANTITY INT NOT NULL CHECK (QUANTITY > 0),
  CUSTOMER_ID INT NOT NULL,
  PRODUCT_ID INT NOT NULL,
  FOREIGN KEY (CUSTOMER_ID) REFERENCES CUSTOMER(CUSTOMER_ID),
  FOREIGN KEY (PRODUCT_ID) REFERENCES PRODUCT(PRODUCT_ID)
);
```

```
Table created.
```



IV. DML (Screenshots of tables along with the values):

1. Customers table:

INSERT INTO CUSTOMERS VALUES (202000846, 'haifa', 'almutairi', 'abullah St, 'haifa00', 'haifamu', '123456', 'alkhobar', 'saudi');

INSERT INTO CUSTOMERS VALUES (202301026, 'najla', 'algarni', 'khaled Road', 'najla321', 'najlaq', '123987', 'aldammam', 'saudi');

CUSTOMER_ID	FIRST_NAME	LAST_NAME	ADDRESS	USER_PASSWORD	USERNAME	ZIP	CITY	STATE
202000846	haifa	almutairi	abullah St	haifa00	haifamu	123456	alkhobar	saudi
202301026	najla	algarni	khaled Road	najla321	najlaq	123987	aldammam	saudi

Download CSV

2 rows selected.

2. **Orders table:**

INSERT INTO ORDERS VALUES (101, '2025-01-10', 299.99, 202301026);

INSERT INTO ORDERS VALUES (102, '2025-01-12', 149.50, 202000846);

ORDER_ID	ORDER_DATE	TOTAL_PRICE	CUSTOMER_ID
101	10-JAN-25	299.99	202301026
102	12-JAN-25	149.5	202000846

Download CSV

2 rows selected.

3. **Payments table:**

INSERT INTO PAYMENT_DETAILS VALUES (9002, '2025-01-12', 'Cash', 'Yes', 0, 202000846);

INSERT INTO PAYMENT_DETAILS VALUES (9001, '2025-01-10', 'Card', 'No', 12345678, 202301026);

PAYMENT_ID	PAYMENT_DATE	PAYMENT_TYPE	CASH	CARD	CUSTOMER_ID
9002	12-JAN-25	Cash	Yes	0	202000846
9001	10-JAN-25	Card	No	12345678	202301026

Download CSV

2 rows selected.

4. **Products table:**

INSERT INTO PRODUCT VALUES (1001, 'WALLETS', 'Accessories', 29.99, 50, 101);

INSERT INTO PRODUCT VALUES (1002, 'HEADBANDS', 'Accessories', 19.99, 100, 101);

INSERT INTO PRODUCT VALUES (1003, 'DRESSES, 'CLOTHING', 99.50, 30, 102);

PRODUCT_ID	PRODUCT_NAME	PRODUCT_CATEGORY	PRODUCT_PRICE	PRODUCT_AVAILABILITY	ORDER_ID
1001	WALLETS	Accessories	29.99	50	101
1002	HEADBANDS	Accessories	19.99	100	101
1003	DRESSES	CLOTHING	99.5	30	102

Download CSV

3 rows selected.

5. **Tracking table:**

INSERT INTO TRACKING_INFO VALUES (501, 'NAQEL, 101);

INSERT INTO TRACKING_INFO VALUES (502, 'SBL', 102);

TRACKING_ID	COURIER	ORDER_ID
501	NAQEL	101
502	SBL	102

Download CSV

2 rows selected.

5. **Website table:**

INSERT INTO WEBSITE (DOMAIN_NAME, CUSTOMER_ID)

VALUES ('HN-CLOTHS', 202000846);

DOMAIN_NAME	CUSTOMER_ID
HN-CLOTHS.com	202000846

Download CSV

6. **Cart table:**

INSERT INTO CART VALUES (7002, 1, 202000846, 1002);

INSERT INTO CART VALUES (7001, 2, 202301026, 1001);

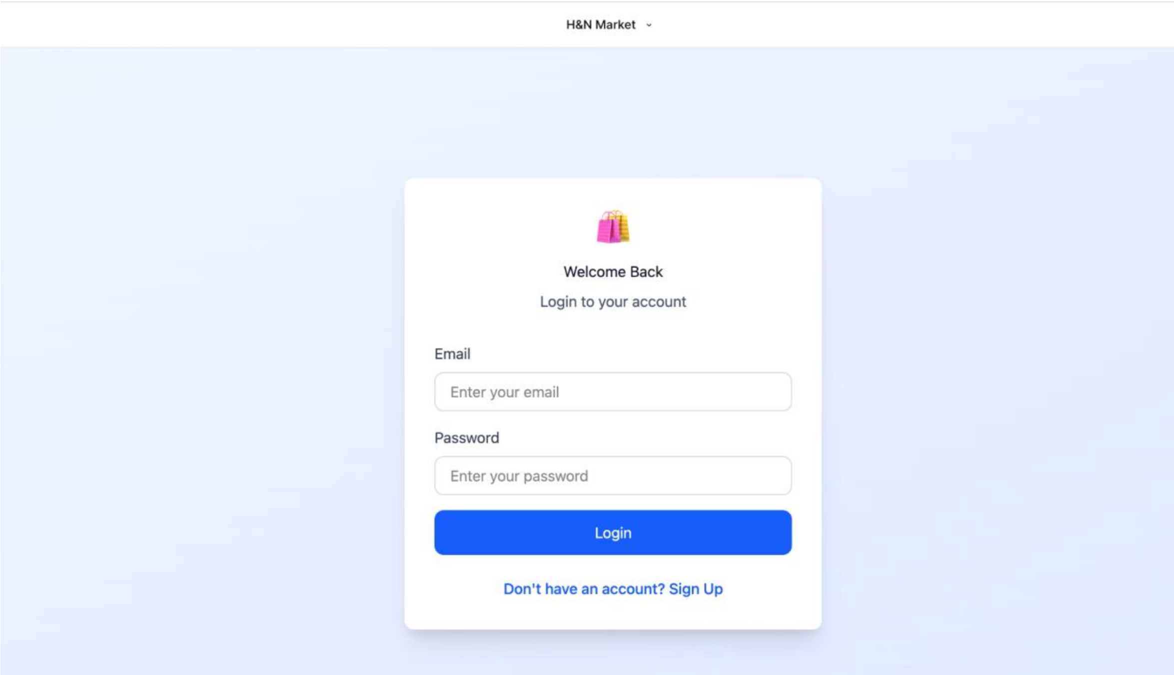
INSERT INTO CART VALUES (7003, 3, 202301026, 1003);

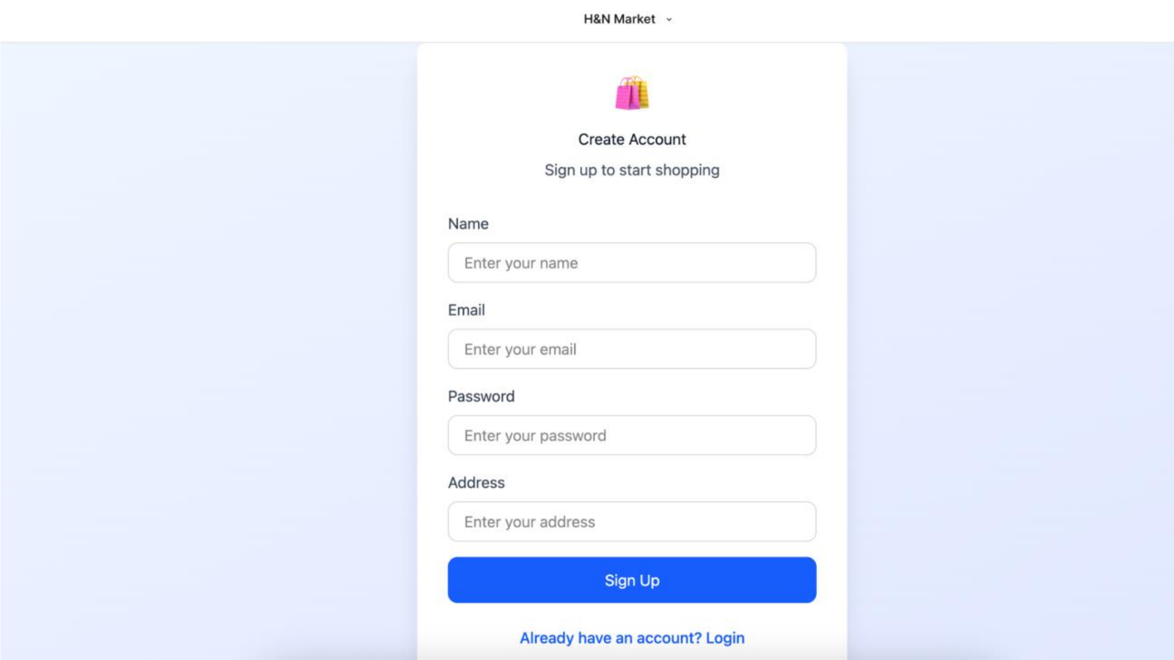
CART_ID	QUANTITY	CUSTOMER_ID	PRODUCT_ID
7002	1	202000846	1002
7001	2	202301026	1001
7003	3	202301026	1003

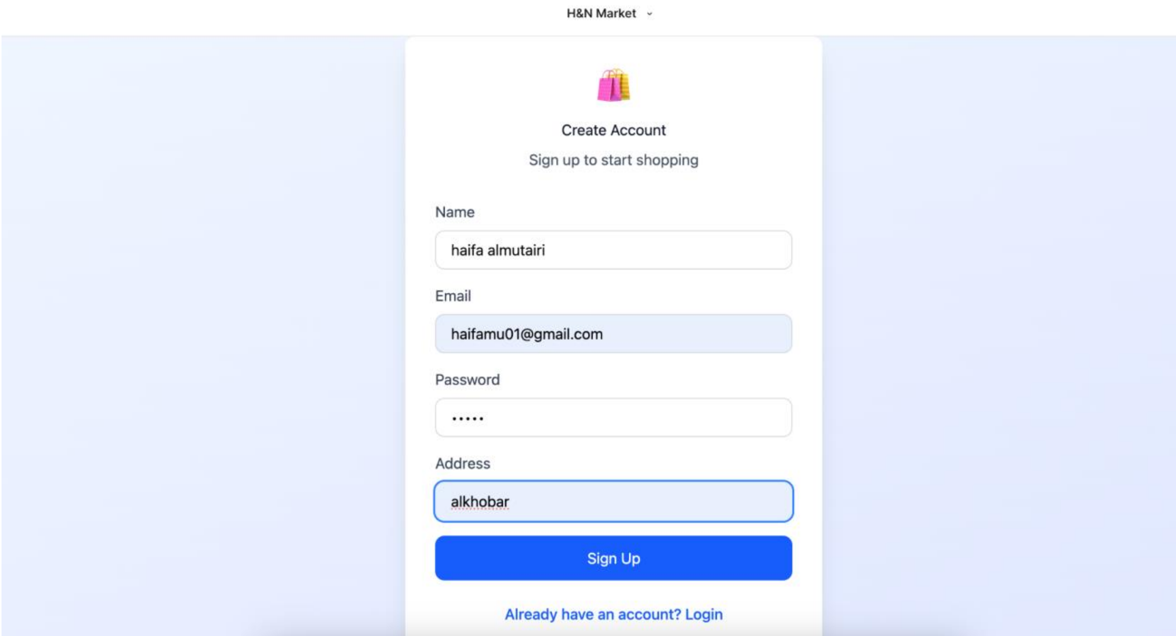
Download CSV

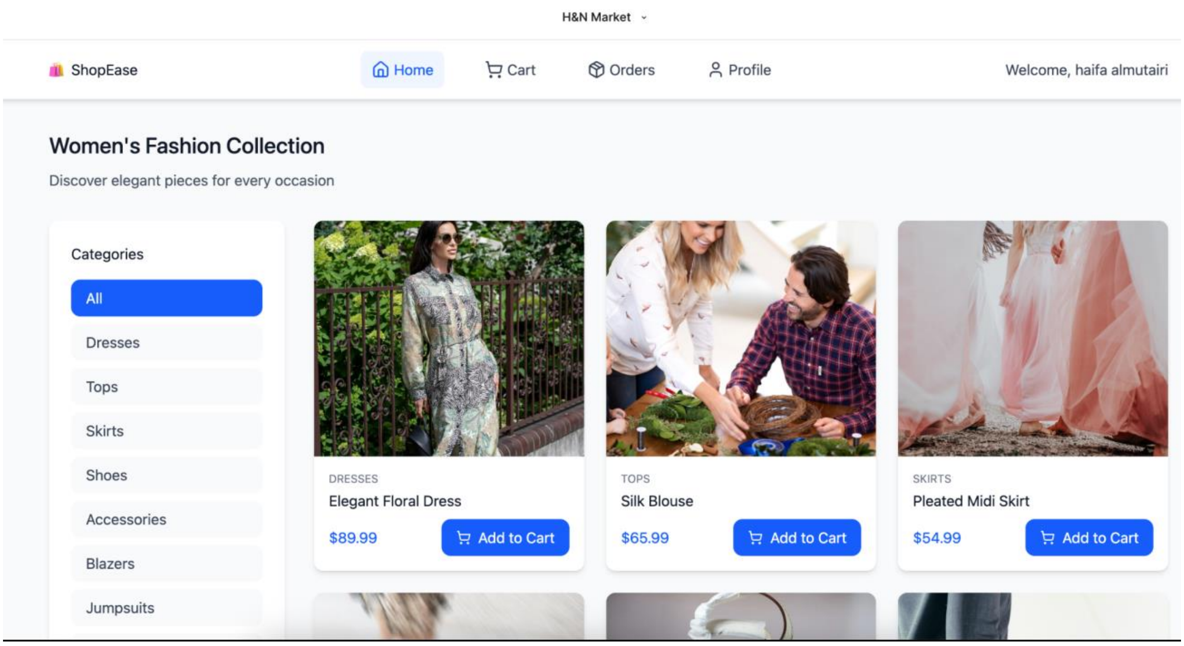
3 rows selected.

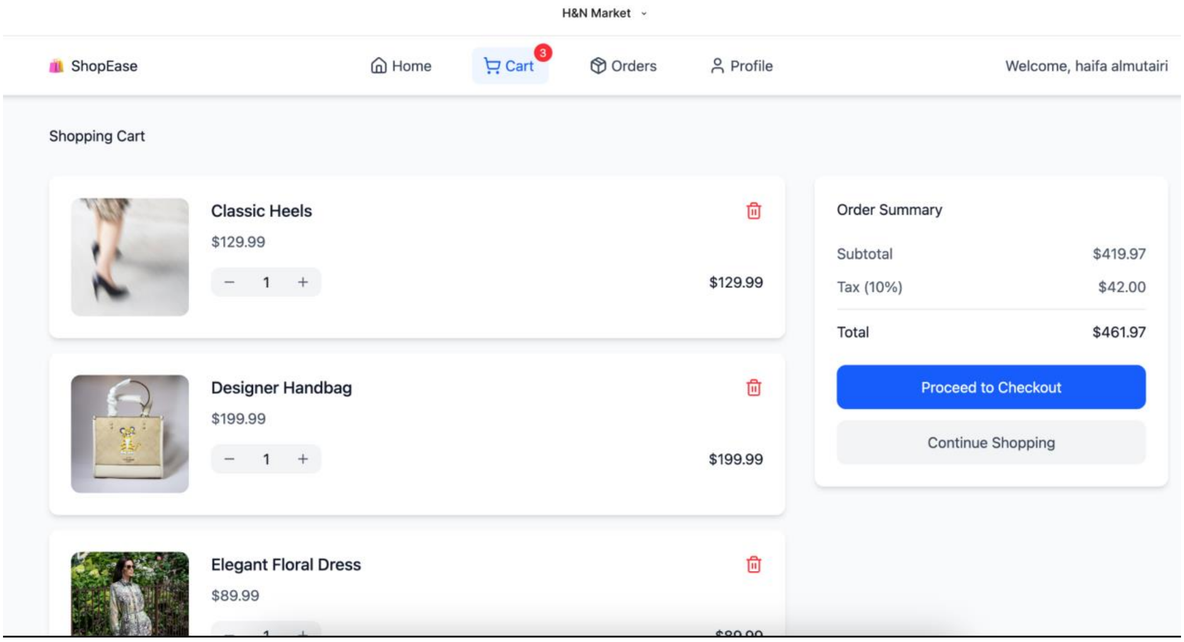
Interface:


H&N Market
Welcome Back
Login to your account
Email
Enter your email
Password
Enter your password
Login
Don't have an account? Sign Up

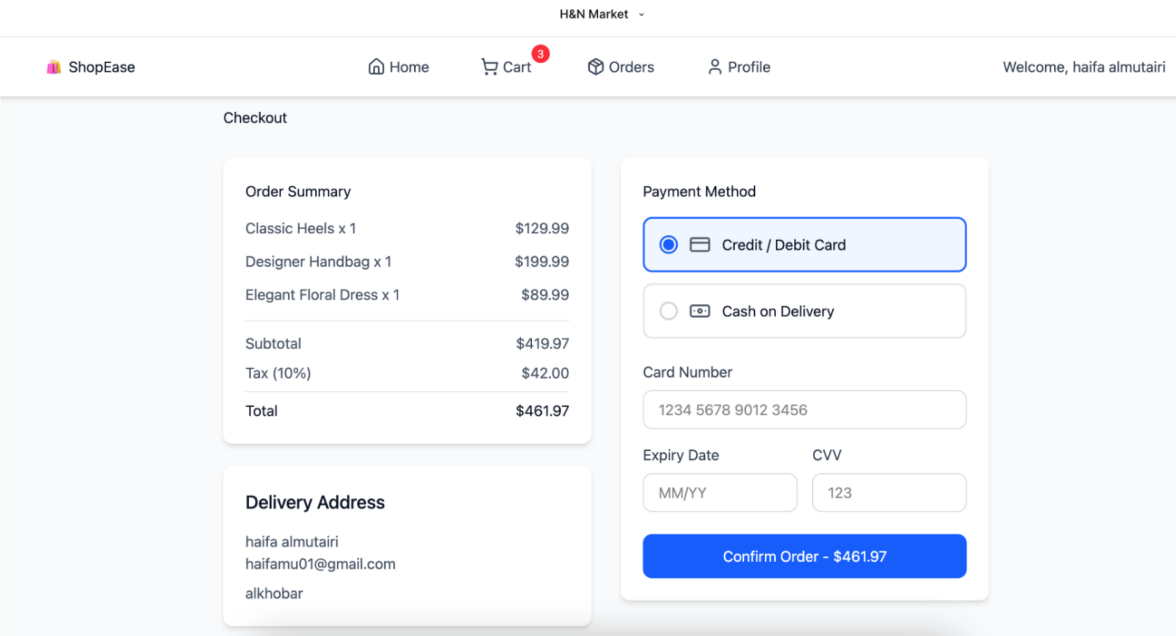

H&N Market
Create Account
Sign up to start shopping
Name
Enter your name
Email
Enter your email
Password
Enter your password
Address
Enter your address
Sign Up
Already have an account? Login


H&N Market
Create Account
Sign up to start shopping
Name
haifa almutairi
Email
haifamu01@gmail.com
Password
Address
alkhobar
Sign Up
Already have an account? Login

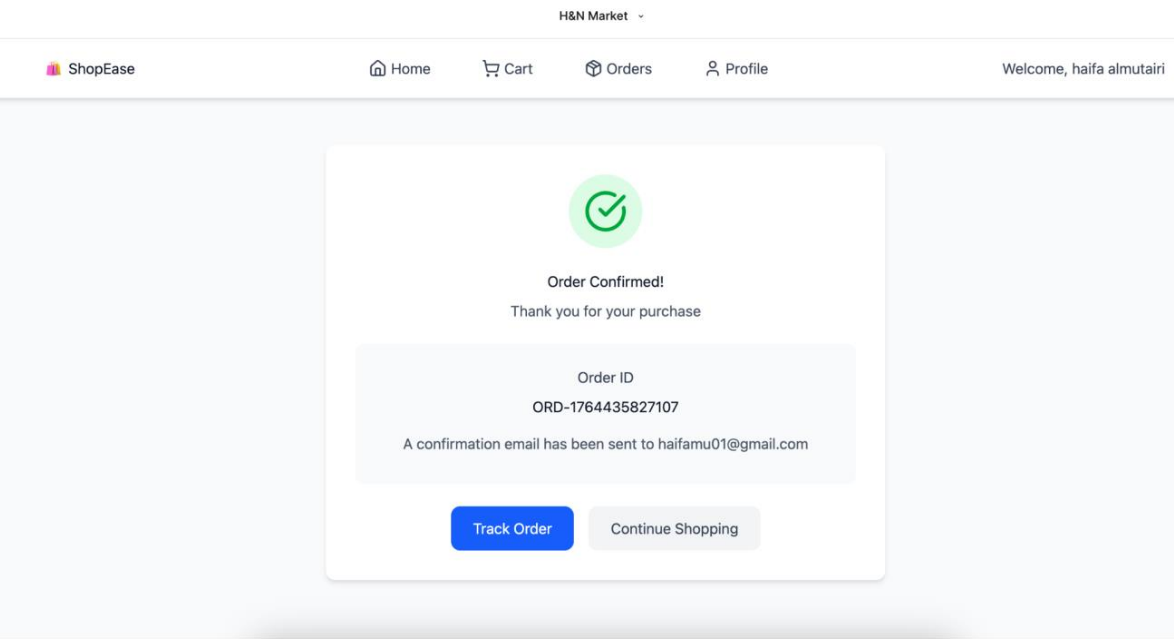

H&N Market
ShopEase
Home
Cart
Orders
Profile
Welcome, haifa almutairi
Women's Fashion Collection
Discover elegant pieces for every occasion
Categories
All
Dresses
Tops
Skirts
Shoes
Accessories
Blazers
Jumpsuits
DRESSES
Elegant Floral Dress
$89.99
Add to Cart
TOPS
Silk Blouse
$65.99
Add to Cart
SKIRTS
Pleated Midi Skirt
$54.99
Add to Cart

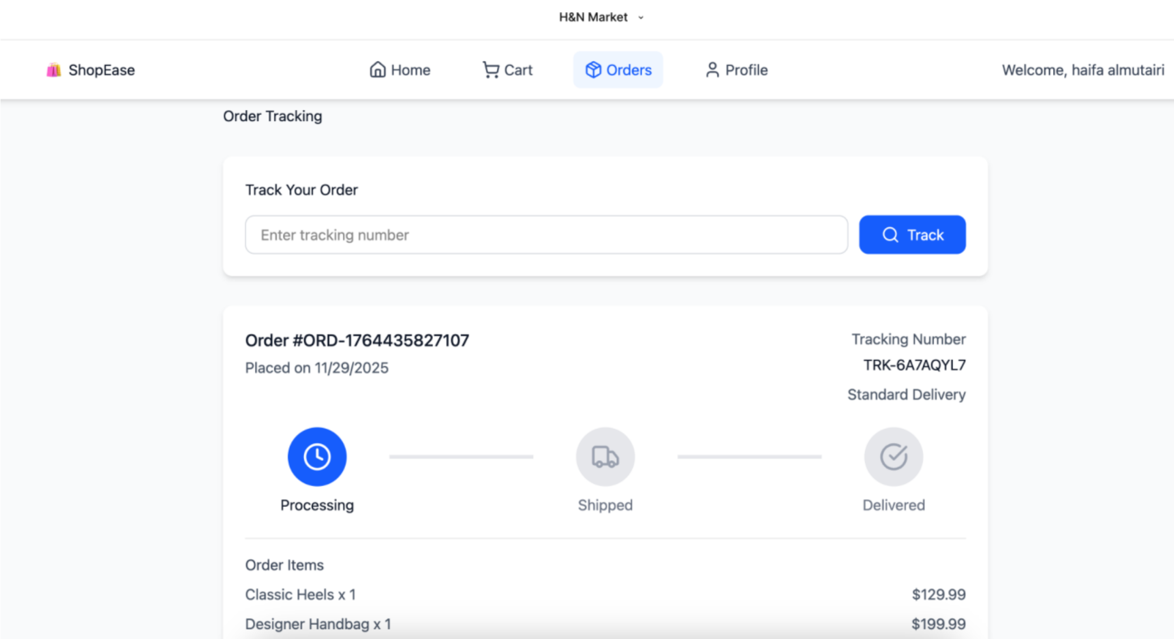


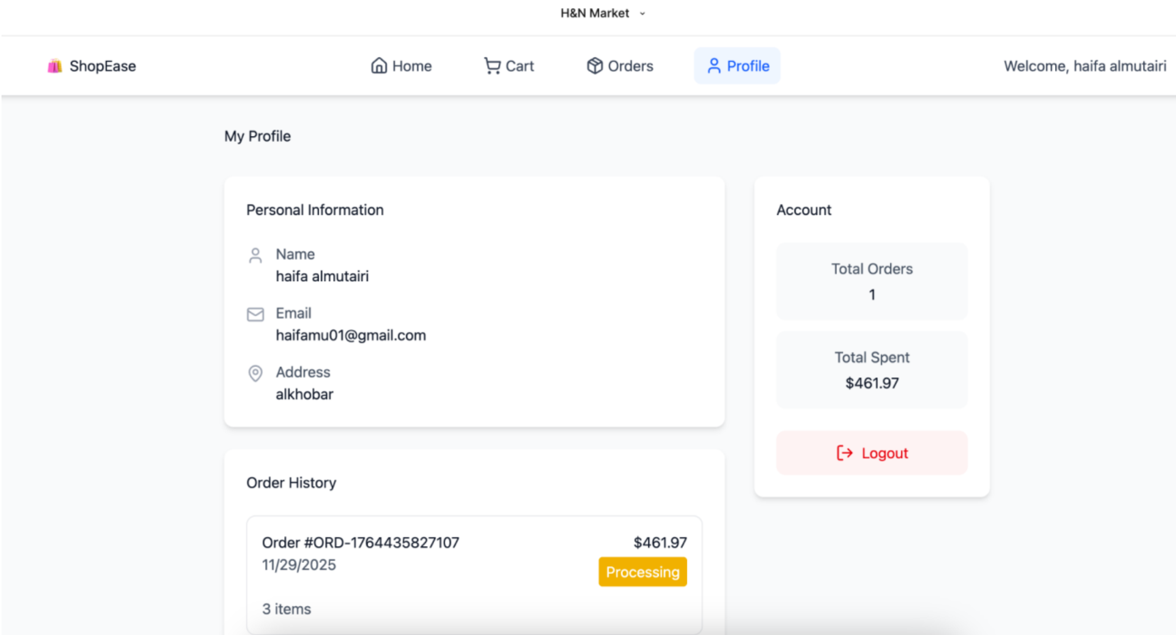
H&N Market
ShopEase
Home
Cart
3
Orders
Profile
Welcome, haifa almutairi
Shopping Cart
Classic Heels
$129.99
1
$129.99
Designer Handbag
$199.99
1
$199.99
Elegant Floral Dress
$89.99
Order Summary
Subtotal $419.97
Tax (10%) $42.00
Total $461.97
Proceed to Checkout
Continue Shopping

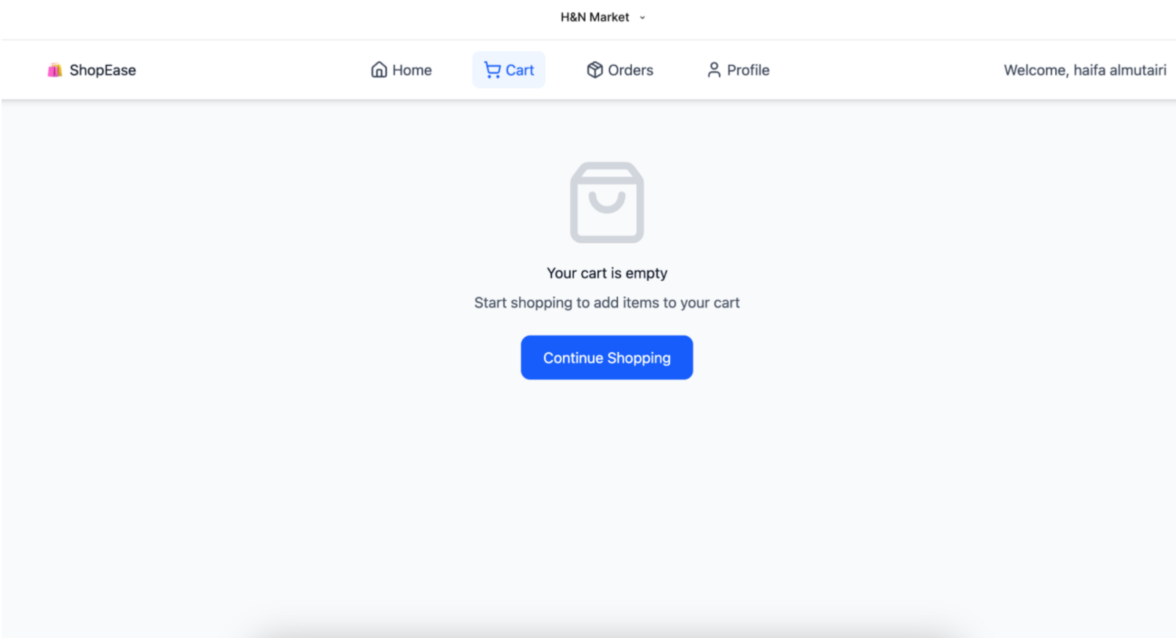


H&N Market
ShopEase
Home
Cart
3
Orders
Profile
Welcome, haifa almutairi
Checkout
Order Summary
Classic Heels x 1 $129.99
Designer Handbag x 1 $199.99
Elegant Floral Dress x 1 $89.99
Subtotal $419.97
Tax (10%) $42.00
Total $461.97
Delivery Address
haifa almutairi
haifamu01@gmail.com
alkhobar
Payment Method
Credit / Debit Card
Cash on Delivery
Card Number
1234 5678 9012 3456
Expiry Date
CVV
MM/YY
123
Confirm Order - $461.97


H&N Market
ShopEase
Home
Cart
Orders
Profile
Welcome, haifa almutairi
Order Confirmed!
Thank you for your purchase
Order ID
ORD-1764435827107
A confirmation email has been sent to haifamu01@gmail.com
Track Order
Continue Shopping


H&N Market
ShopEase
Home
Cart
Orders
Profile
Welcome, haifa almutairi
Order Tracking
Track Your Order
Enter tracking number
Track
Order #ORD-1764435827107
Placed on 11/29/2025
Tracking Number
TRK-6A7AQYL7
Standard Delivery
Processing
Shipped
Delivered
Order Items
Classic Heels x 1
$129.99
Designer Handbag x 1
$199.99


H&N Market
ShopEase
Home
Cart
Orders
Profile
Welcome, haifa almutairi
My Profile
Personal Information
Name
haifa almutairi
Email
haifamu01@gmail.com
Address
alkhobar
Account
Total Orders
1
Total Spent
$461.97
Logout
Order History
Order #ORD-1764435827107
$461.97
11/29/2025
Processing
3 items


H&N Market
ShopEase
Home
Cart
Orders
Profile
Welcome, haifa almutairi
Your cart is empty
Start shopping to add items to your cart
Continue Shopping